1 Introduction

If $x^2 + 5x$ is odd than x is odd.

Proof. Contrapositive: if x is even than $x^2 + 5x$ is even

Lemma 1.1. *x is even if and only if $\exists K \in Z, 2K = x$*

Since $x = 2k$, we substitute x for $2k$ in $x^2 + 5x$
$(2k)^2 + 5(2k)$ $4k^2 + 10k$
We can factor out a 2 from this expression
$2(2k^2 + 5k)$
The expression being multiplied by 2 is an integer. Therefore, since x is even (2k), $x^2 + 5x$ is even.
QED the contrapositive is true so the original claim is also true.

□

proof
Given $x_1 < x_2$, if $f(x_1) > f(x_2)$ then $f(x)$ is not increasing

Lemma 1.2. *$f(x)$ is increasing*